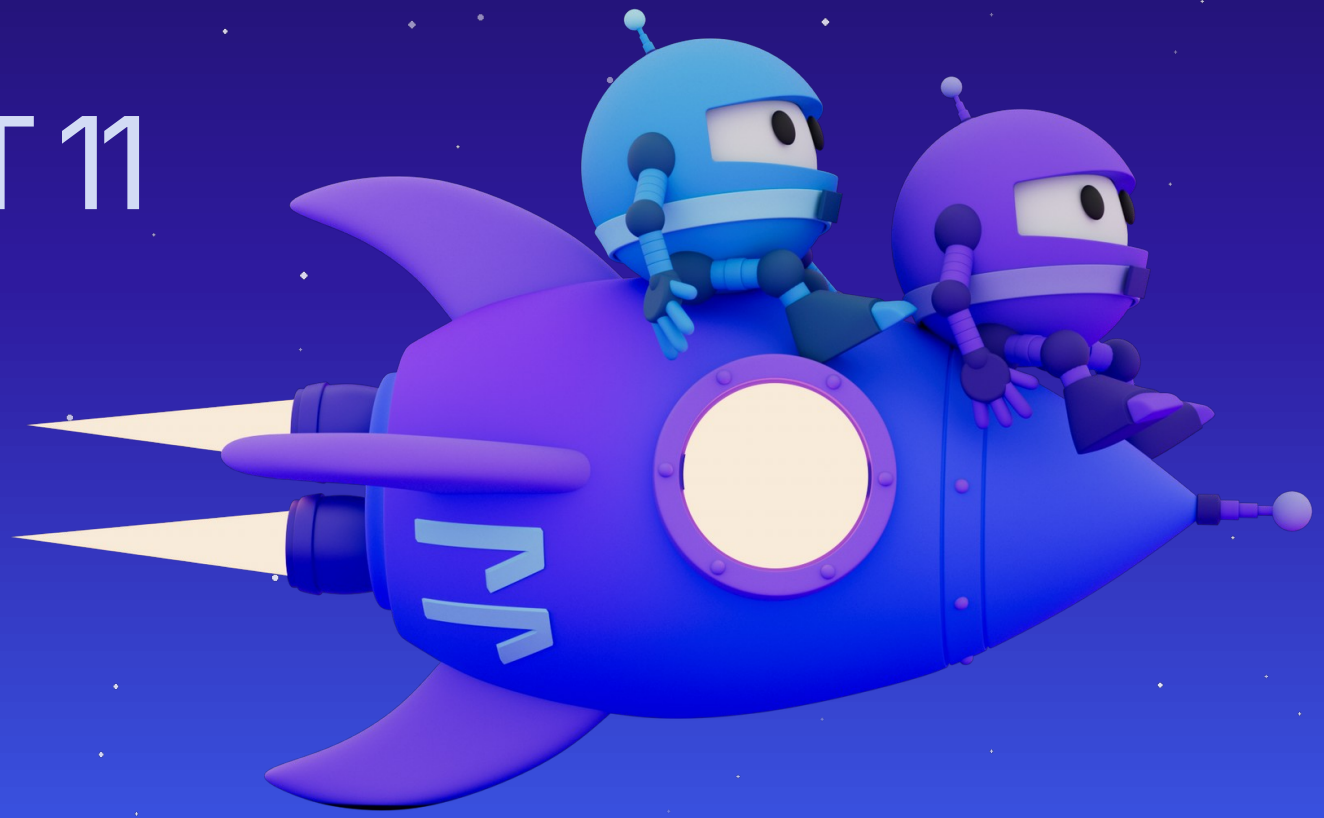
Infrastructure as (Actual) Code

with C# 15 and .NET 11

.NET



Cam Soper

Systems engineering, Pulumi

Infrastructure as code, they said

```
resource "azurerm_storage_account" "site" {
  name                     = "site${var.env}"
  resource_group_name      = "rg-${var.env}"
  location                 = "centralus"
  account_tier             = "Standard"
  account_replication_type = var.env == "prod" ? "ZRS" : "LRS"
}

resource "azurerm_management_lock" "keep" {
  count      = var.env == "prod" ? 1 : 0
  name       = "do-not-delete"
  scope      = azurerm_storage_account.site.id
  lock_level = "CanNotDelete"
}
```

That's not code.
That's config with
extra steps.



Cam Soper

Systems engineering, Pulumi

13 years at Microsoft: .NET and Azure, co-host of On .NET Live

At Pulumi: built our automated docs QA system. Now: our agentic SDLC

What “actual code” gets you

Things C# devs already have

A compiler

`error CS8509`

Types and IntelliSense

`SkuName.Standard_ZRS`

Tests

`dotnet test`

Packages

`dotnet add package`

Pulumi, in one slide

Declare what you want

Pulumi diffs and deploys

Open source, no account:
`pulumi login --local`

```
var group = new ResourceGroup("demo");

var storage = new StorageAccount("site",
    new()
    {
        ResourceGroupName = group.Name,
        Kind = Kind.StorageV2,
        Sku = new SkuArgs
        {
            Name = SkuName.Standard_LRS,
        },
    });
```

01 Demo: Hello, cloud



C# 15: union types

Exactly one of a fixed set

Miss a case: compiler error

No default arm needed

```
public record Dev;
public record Staging;
public record Prod(string OnCall);

public union Stage(Dev, Staging, Prod);

SkuName Redundancy(Stage s) => s switch
{
    Dev     => SkuName.Standard_LRS,
    Staging => SkuName.Standard_LRS,
    Prod    => SkuName.Standard_ZRS,
};
```

02 Demo: The compiler is your policy engine



I wrote a rule for my agents. In bold.

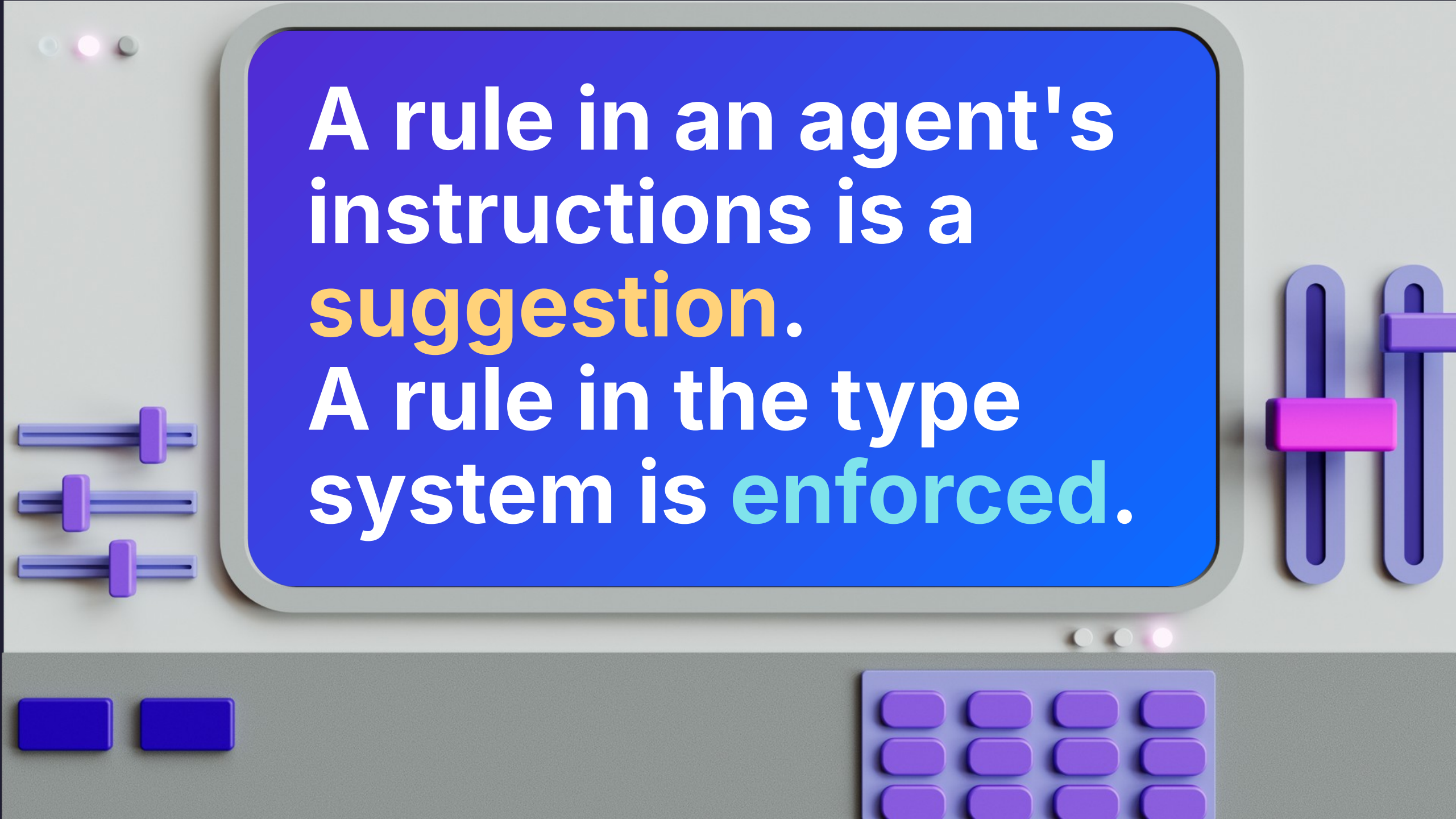
From my own CLAUDE.md:

“Do NOT put Access in front of hades: it self-implements OAuth and would double-prompt.”

```
public union McpAuth(
    SelfOAuth,
    AccessEdgeOnly,
    AccessWithOriginCheck);

record McpService(
    string Name, McpAuth Auth);

var hades = new McpService(
    "hades", new SelfOAuth());
// One service. One posture.
```


A rule in an agent's instructions is a suggestion.
A rule in the type system is enforced.

03 Demo: Agent vs. compiler



Take it home
Slides, code, and notes

`demos/`

Three runnable demos

`tests/`

The whole stack vs. a fake cloud. No Azure.

`bonus/single-file/deploy.cs`

One file. dotnet run.

Gotcha

```
#:property PublishAot=false
```



github.com/CamSoper/infrastructure-as-actual-code

Thank you

github.com/CamSoper/infrastructure-as-actual-code

pulumi.com/docs/iac/languages-sdks/dotnet

“What's new in C# 15” on Microsoft Learn

